

Translated

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Z-function and its calculation

Suppose we are given a string s of length n . The **Z-function** for this string is an array of length n where the i -th element is equal to the greatest number of characters starting from the position i that coincide with the first characters of s .

In other words, $z[i]$ is the length of the longest string that is, at the same time, a prefix of s and a prefix of the suffix of s starting at i .

Note. In this article, to avoid ambiguity, we assume 0-based indexes; that is: the first character of s has index 0 and the last one has index $n-1$.

The first element of Z-function, $z[0]$, is generally not well defined. In this article we will assume it is zero (although it doesn't change anything in the algorithm implementation).

This article presents an algorithm for calculating the Z-function in $O(n)$ time, as well as various of its applications.

Examples

For example, here are the values of the Z-function computed for different strings:

- "aaaaa" - $[0, 4, 3, 2, 1]$
- "aaabaab" - $[0, 2, 1, 0, 2, 1, 0]$
- "abacaba" - $[0, 0, 1, 0, 3, 0, 1]$

Trivial algorithm

Formal definition can be represented in the following elementary $O(n^2)$ implementation.

```
vector<int> z_function_trivial(string s) {  
    int n = s.size();  
    vector<int> z(n);  
    for (int i = 1; i < n; i++) {  
        while (i + z[i] < n && s[z[i]] == s[i + z[i]]) {  
            z[i]++;  
        }  
    }  
}
```